vllm-sr/d3

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d3

d3 is the 27B multimodal foundation decision model of Decision 3.0, the decision models of vLLM Semantic Router. Give it an input (text or JSON, optionally with images and videos) and the questions you need answered: pick one of several options, say yes or no, or rate on a scale. It answers them all in one call and returns a probability for every answer, without generating text.

Downloads last month
1,156

Safetensors
Model size 26B params
Tensor type BF16 | Chat template
Files info

Inference Providers NEW
Zero-Shot Classification
This model isn't deployed by any Inference Provider.
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Model tree for vllm-sr/d3
Base model Qwen/Qwen3.8-27B
Finetuned (532) this model
Quantizations 8 models

Spaces using vllm-sr/d3 5
vllm-sr/d3

Parameters	26.09B, including the 0.46B vision encoder
Inputs	Text or JSON, plus images and videos (several per request)
Decision types	Choice · Yes / No · Score
License	Apache-2.0

Highlights

- **Jev Decision Index 0.3.2: 66.3, first on the official text board** and on its Pareto frontier; public suite 65.45, measured with the official 0.3 kit on the released weights: all 140,178 public requests answered, none unsupported.
- **+10.4 over Decision 2.0** (its 27B model: 55.9 on the official board): +8.5 on the public suite, ahead in all five of its areas, and +7.1 and +12.6 on the private same-skill tests and new-domain tasks.
- **Reads images:** multiple images per request (PNG, JPEG or WebP), given as paths, URLs, PIL images or base64 data URLs; every question of the request sees all of them.
- **Reads videos:** multiple videos per request (MP4, WebM, MOV or MKV), given as paths, URLs, base64 data URLs or frame arrays, read at 2 frames per second; **Perception Test (multiple-choice video QA): 77.4** (internal evaluation).
- **Speed:** text requests take a median of 54.4 ms, requests with an image a median of 290 ms, and

vllm-sr/d3-decision-model

multimodalart/jev-decision-index

tardellirs/model-pulse

emanforhelp/model-pulse

Collection including `vllm-sr/d3`

Decision 3.0 Collection

Towards O… • 7 items • Upd… • △ 97

requests with a 10-second video a median of 500 ms, on one AMD Instinct MI325X, one request at a time with AutoModel.

- **Serving:** 27 requests/s with p99 under 1 s on one AMD Instinct MI325X with vLLM Semantic Router, 1.5× AutoModel (one-question text, `batching` profile); the default `exact` profile gives AutoModel's answers byte for byte when both use the same kernel settings (Serving performance).
- **Many questions, one call:** Choice, Yes / No and Score questions about the same input are answered together, each from its own forward pass over the input, with a probability for every option.

Quickstart

```
pip install "transformers==5.17.0" torch tor
pip install flash-linear-attention  # option
```

```
import json

from huggingface_hub import hf_hub_download
from transformers import AutoModel

model = AutoModel.from_pretrained("vllm-sr/d

# Text
result = model.system_one(
    state="The order arrived damaged yesterda
    questions={
        "route": {
            "type": "choice",
            "instructions": "Which team shou
            "criteria": {
```

```
                "returns": "Refunds, replacer
                "billing": "Payments, invoic
                "technical": "Product setup a
            }
        },
        "receipt": {
            "type": "noul",
            "instructions": "Does the custom
        },
        "urgency": {
            "type": "score",
            "instructions": "How urgent is th
            "criteria": [
                "Routine",
                "Soon",
                "Today"
            ]
        }
    },
)
print(json.dumps(result["answers"], indent=2)

# Text and an image (or several)
receipt = hf_hub_download("vllm-sr/d3", "assc
result = model.system_one(
    state="The customer says the blender arri
    images=[receipt],  # local paths, http(s
    questions={
        "route": {
            "type": "choice",
            "instructions": "Which team shou
            "criteria": {
                "returns": "Refunds, replacer
                "billing": "Payments, invoic
                "technical": "Product setup a
            }
        },
        "on_receipt": {
            "type": "noul",
            "instructions": "Does the receip
        },
```

```
        "payment": {
            "type": "choice",
            "instructions": "How was the ord
            "criteria": {
                "card": None,
                "cash": None,
                "gift card": None
            }
        }
    },
)
print(json.dumps(result["answers"], indent=2

# Text and a video (or several)
clip = hf_hub_download("vllm-sr/d3", "assets
result = model.system_one(
    state="A short clip from a test camera."
    videos=[clip],  # local paths, http(s) U
    questions={
        "direction": {
            "type": "choice",
            "instructions": "Which way does
            "criteria": {
                "right": "From left to right
                "left": "From right to left"
                "still": "It does not move"
            }
        },
        "color_change": {
            "type": "noul",
            "instructions": "Does the square
        }
    },
)
print(json.dumps(result["answers"], indent=2

# Or as a pipeline:
# transformers.pipeline("decision", model="v
```

Images go before the text of the request, each read at up to 1.6 megapixels; videos follow the images,

read at 2 frames per second (at most 32 frames spread over the whole video, each at up to 0.2 megapixels). Every question of the request sees all of them.

Run with vLLM Semantic Router

vLLM Semantic Router serves Decision 3.0 natively in Engine mode, which runs one model behind the System One API on your own machine. Its answers match this package's own `d3_server.py` exactly, for text, images and videos (measured on AMD Instinct MI325X).

```
curl -fsSL https://vllm-sr.ai/install.sh | ba
export PATH="$HOME/.local/bin:$PATH"
vllm-sr serve -e vllm-sr/d3
```

The platform is detected automatically; choose one with `--platform auto|cpu|cuda|rocm` (default `auto`).

Engine mode and Decision 3.0 are in the development channel (`--channel dev`) until a stable vLLM Semantic Router release includes them. The first start pulls the image for your machine and downloads the model. Then ask questions about a request:

```
curl -s localhost:8899/v1/systemone -H 'conte
  "model": "vllm-sr/d3",
  "state": "Write a Python function that merg
  "questions": {
    "kind": {"type": "choice", "instructions
             "criteria": {"code": "Writing o
    "reasoning": {"type": "noul", "instructi
```

```
    }
  }'
```

Send images as base64 data URLs (PNG, JPEG or WebP) in `images` and videos (MP4, WebM, QuickTime or Matroska) in `videos`; every question sees them:

```
curl -s localhost:8899/v1/systemone -H 'conte
{
  "model": "vllm-sr/d3",
  "state": "What does the attached image sho
  "images": ["$(base64
  "questions": {
    "kind": {"type": "choice", "instructions
             "criteria": {"chart": "A chart
  }
}
EOF
```

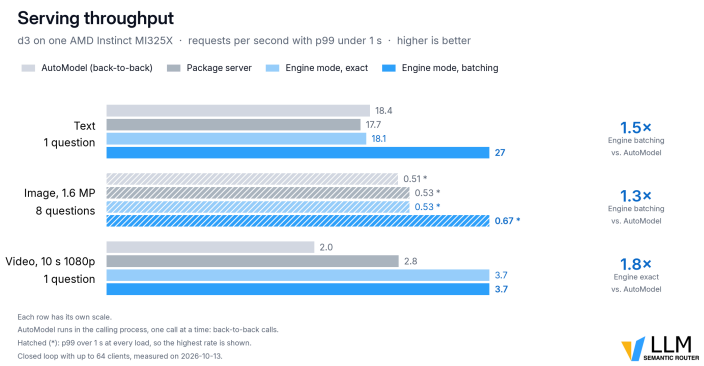
```
curl -s localhost:8899/v1/systemone -H 'conte
{
  "model": "vllm-sr/d3",
  "state": "What happens in the clip?",
  "videos": ["data:video/mp4; base64,$(base64
  "questions": {
    "moving": {"type": "noul", "instructions
  }
}
EOF
```

`vllm-sr stop` stops it. See Router and Engine modes for limits and options.

Serving performance

On one AMD Instinct MI325X, vLLM Semantic Router serves d3 at 27 requests/s with p99 under 1 s, 1.5×

AutoModel (one-question text, `batching` profile), and its default `exact` profile gives AutoModel's answers byte for byte when both use the same kernel settings.



AutoModel runs in your own process, one call at a time (the Quickstart code); the package server is its `python d3_server.py`; Engine mode is `vllm-sr serve -e` with its default `exact` profile, or with `--runtime-profile batching`.

Where Engine mode helps, and what it costs:

- **Throughput:** 1.5× AutoModel's requests per second on one-question text, 1.3× on 8-question images and 1.8× on 10-second videos, with the better Engine profile of each row (8-question images: highest rates, since no side keeps p99 under 1 s there).
- **Start-up:** the first answer comes 19.4 s after launch (20.4 s with `batching`) with empty kernel caches, against 38.4 s for AutoModel, whose first image then takes another 6.20 s, and 80.8 s for the package server.
- **Many questions and long inputs:** one request at a time, Engine mode (`exact`) answers 16 text questions in 548 ms (AutoModel: 572 ms), 8 questions about one image in 1.87 s (1.95 s) and

a 32k-token text in 4.21 s (4.23 s); `batching` runs the 8 image questions together in 1.56 s, 20 % less than AutoModel.

- **GPU memory:** 65.4 GiB per replica at peak with `exact`, about the same as the package server's 65.2 GiB, and 62.1 GiB with `batching`; AutoModel, serving one request at a time, peaked at 61.8 GiB. Engine mode keeps up to 2 GiB of image features cached.
- **Answers:** Engine mode (`exact`) returned answers byte-identical to AutoModel's on all 7 request types (text with 1, 4 or 16 questions, an image with 1 or 8, the video and a 32k-token text) when AutoModel used the same kernel settings: 135 Engine replies and 21 AutoModel calls compared. With the kernel autotuning AutoModel runs in each new process, 5 of the 7 were byte-identical and no decision changed (largest probability difference 0.005). `batching` changed 0.048 % of decisions on a 4,000-request sample of the public suite (14 of 29,339 questions, near-ties), with gold accuracy 92.109 → 92.126 %.

A single short request is fastest in your own process: AutoModel answers one text question 0.9 ms sooner than Engine mode and one new image 22 ms sooner, the cost of going through the router to the model worker and its media path. Engine mode's gains come from serving many requests at once.

▶ Single-request latency and the numbers behind the chart

Measured on 2026-10-13, one AMD Instinct MI325X per replica and each side alone on the GPU: ROCm 7.2 with PyTorch 2.12; AutoModel

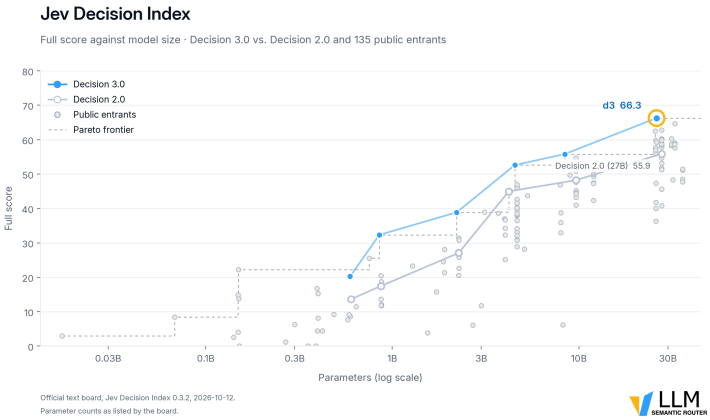
and the package server with Transformers 5.17.0; vLLM Semantic Router Engine mode built from source, with serving optimizations that are newer than its published images. Requests: a support ticket of about 330 tokens with 1, 4 or 16 questions; a 1,456 × 1,092 PNG (1.6 MP) with 1 or 8 questions; a 10-second 1080p MP4, read at 2 frames per second; a 32k-token text; every image and video request carries new bytes. One request at a time: 3 warm-up calls, then up to 20 timed calls. Under load: 1, 2, 4, … 64 closed-loop clients, each level warmed up and then counted for 15 to 60 s; Engine mode admits 32 requests in flight per worker and answers the rest at once with 503. Start-up: model files on local disk; Engine mode from the start of its router process, without the CLI's image pull.

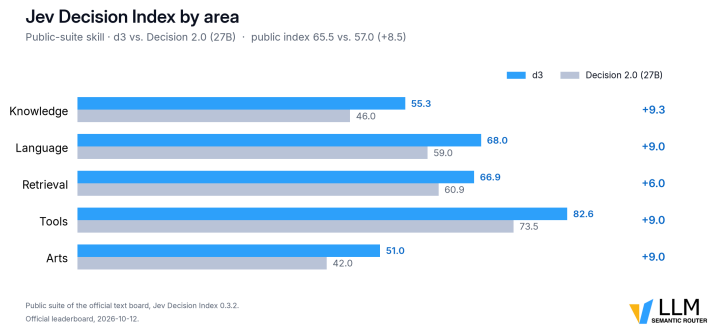
Evaluation

Text: Jev Decision Index 0.3.2

Model	Jev Decision Index ↑	Public ↑	Same-skill tests ↑	New-domain tasks ↑
d3	**66.3**	**65.5**	**62.8**	**60.5**
Decider 31B	64.6	64.3	61.8	58.3
Perplexity Decider v1.1 (27B)	62.8	62.3	61.1	55.6
Darwin-27B-ZTC	62.5	62.6	58.7	57.7
AnyJev L0 · Gemma-4-31B	60.2	59.8	59.1	53.0
Decision 2.0 (27B)	55.9	57.0	55.7	47.9

d3 ranks first of the 149 entries on the official text board and is on its Pareto frontier: no entry with as many parameters or fewer scores higher. Five of the six Decision 3.0 models are on that frontier; d3-edge, second among the entries of its size, is not.





Images: Jev Decision Index vision board 0.3.2

Model	Vision Index ↑	Public ↑	Private ↑
d3	**71.3**	**74.1**	**68.5**
Perplexity Decider v1.1 (27B)	70.6	73.2	67.9
JEV-27B-VL	69.6	72.8	66.4
Solomon v1.1 (27B)	66.8	69.9	63.8

On the official vision board, the six Decision 3.0 models form the whole Pareto frontier: no other

entry is on it. d3 ranks first of the 33 entries.

d3 on the public vision benchmarks († approximate rebuild):

Benchmark	d3
CV-Bench	76.2
BLINK	59.9
RealWorldQA	73.4
CharXiv †	91.7
InfographicVQA †	96.7
Mind2Web †	90.0
Winoground	85.8
KIE (CORD+FUNSD) †	97.7
Moderation (Hateful Memes)	43.9
R-Bench-M	33.0
MMMU-Pro vision	47.0

Videos: Perception Test

Perception Test (validation)	d3
All 19,140 questions	**77.4**
Memory	80.3
Abstraction	71.2
Physics	72.5
Semantics	88.8

Multiple-choice video question answering (three options per question, chance 33.3): top-1 accuracy (%), videos read with the defaults above.

Jev Decision Index: official leaderboard, text and vision boards 0.3.2, 2026-10-12; the boards evaluated v3.1.0, which has the same weights as this version. Size bands and the Pareto frontier use the parameter counts listed by the boards. d3 on the public vision benchmarks and the Perception Test: internal evaluation.

License

Apache-2.0 (LICENSE). Built on Qwen/Qwen3.8-27B (Apache-2.0).

Citation

```
@misc{d3_2026,
  title        = {{d3}: A Multimodal Foundat
  author       = {{vLLM Semantic Router Team
  year         = {2026},
  howpublished = {\url{https://huggingface.c
}
```

Trained on AMD Instinct MI325X GPUs.

System theme TOS Privacy About Careers Models Datasets Spaces Pricing Docs